Lecture plan · for the lecturer

Teaching AI security at university: unit check

AI security · University · Unit check

Time	Materials
40 min	Projector or board, and the lecture slides.

Learning goals

- Adversarial examples
- Poisoning and backdoors
- Privacy attacks
- Prompt injection

Activity: Peer-instruction review

1. Vote (10 min): Show each question; students answer alone and commit to an answer.
2. Discuss (15 min): Where answers split, pairs argue for two minutes and vote again.
3. Explain (10 min): Work through the questions most students missed, showing the reasoning step by step.
4. Plan (5 min): Note which misconception to revisit next week and how.

Discussion

- Which misconception will you target first?
- Which lab will you add?
- How will you keep red teaming safe?

Exit ticket

- One change to my AI security course this term, and how I will know it worked.

Try it in your next lecture
Bring your attack-the-defence assignment and red-team scope document to your next department meeting for review.

Seminar sheet · for students

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Name _____ Class _____ Date _____

1 **FGSM moves each feature by ε in the direction of**

◯ The sign of the loss gradient ◯ A random sign ◯ The negative of the input

2 **Bound the change.**

w = (3, -1, 2), ε = 0.1: the score can change by _____ .

Word bank: 0.6 · 0.4 · 0.1

3 **A backdoored model usually shows low clean accuracy.**

◯ True ◯ False

4 **The strongest evidence that a defence is robust:**

◯ It survives adaptive attacks under a stated threat model ◯ FGSM fails against it

◯ High clean accuracy

5 **Set the noise.**

Sensitivity 1, ε = 0.25: Laplace scale = _____ .

Word bank: 4 · 0.25 · 2